# Technical specification template: UPS Batteries

*OHMETRY 2 MW DATA CENTRE ELECTRICAL ENGINEERING PROJECT. Package revision A6.*
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**DRAFT TEMPLATE. NOT AN ISSUED SPECIFICATION. NOT FOR PROCUREMENT. PRELIMINARY VALUES ARE NOT MANUFACTURER-CONFIRMED.**

## Document control

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## 1. Scope

This template defines the technical requirements for ups batteries for the OHMETRY 2 MW DATA CENTRE ELECTRICAL ENGINEERING PROJECT. It is to be completed and issued by the project. Items in [square brackets] are decisions or data the project must supply. Preliminary values from the Ohmetry reference configuration are shown with their data status and must be replaced or confirmed.

## 2. Applicable standards

- IEC 60896 series (stationary lead-acid), if lead-acid is selected
- IEC 62619 (secondary lithium cells and batteries, safety), if lithium-ion is selected
- IEC 62485-2 (safety requirements for stationary batteries)
- IEEE 485 (sizing lead-acid batteries) as a method reference
- Local fire and ventilation regulations [project to add]
- Editions and applicability of every standard above are NOT VERIFIED. The project shall confirm the current editions and any national deviations.

## 3. Service conditions and system data

| Item | Requirement or preliminary value | Data status |
|---|---|---|
| Nominal LV system | 400 V line-to-line, 60 Hz, 3-phase | PRELIMINARY_ASSUMPTION. Utility confirmation is the owner's statement, not independently verified |
| MV supply | 13.8 kV | PRELIMINARY_ASSUMPTION. To be confirmed with the utility |
| Design ambient temperature | 30 degC | PRELIMINARY_ASSUMPTION. Site data to be confirmed |
| Site altitude | 0 m | PRELIMINARY_ASSUMPTION. Site data to be confirmed |
| Prospective fault level | [To be calculated: short-circuit study pending utility, transformer and generator data] | NOT CALCULATED |

## 4. Equipment ratings

Ratings below are preliminary values from the master configuration unless marked otherwise. They are not manufacturer-confirmed and are not an equipment selection.

| Item | Requirement or preliminary value | Data status and note |
|---|---|---|
| Autonomy to be provided | 10 min at the critical load | PRELIMINARY_ASSUMPTION |
| Energy at the UPS output | 333 kWh (arithmetic, not a battery size) | Preliminary |
| DC bus voltage, cell count, strings | [To be determined by the project] | Pending manufacturer constant-power discharge data |
| Chemistry and design life | [Project to select] | Project decision |
| End-of-life capacity, temperature and design margin factors | [To be determined by the project] | Pending manufacturer data |

## 5. Technical requirements

5.1 The battery shall deliver the stated autonomy at the stated constant power to the end-of-discharge voltage at end of life and at the minimum room temperature.

5.2 The manufacturer shall provide the constant-power discharge table (power per cell against time to the end voltage) for the offered cell.

5.3 The manufacturer shall state ageing, temperature correction and expected life and shall declare the capacity at the design discharge rate.

5.4 Battery management, monitoring, protection, racks, seismic requirements and ventilation or gas management: [project to define].

5.5 Thermal runaway and fire safety measures: [project to define with the authority].

## 6. Vendor data schedule

The tenderer shall complete the schedule below with its offer: the value offered, its unit, and the name and revision of the document that states it. A value without a source document is not accepted. The parameters are those in the manufacturer-data register; supplied values are imported into the calculations through the data override mechanism, and a value stays NOT VERIFIED until a reviewer records an attestation against its source (name, organisation, date, evidence reference, scope).

| Parameter id | Parameter | Datasheet required | Current status | Value offered | Unit | Document and revision |
|---|---|---|---|---|---|---|
| battery.dc_bus_voltage_v | DC bus voltage | Battery data sheet with the constant-power discharge table (W per cell to the end voltage at the | MISSING | [vendor to complete] | [vendor to complete] | [vendor to complete] |

| | | | | | | |
|---|---|---|---|---|---|---|
| | | autonomy time), temperature and ageing data | | | | |
| battery.inverter_efficiency_pct | Inverter efficiency on battery | Battery data sheet with the constant-power discharge table (W per cell to the end voltage at the autonomy time), temperature and ageing data | MISSING | [vendor to complete] | [vendor to complete] | [vendor to complete] |
| battery.end_of_life_capacity_factor | End-of-life capacity factor | Battery data sheet with the constant-power discharge table (W per cell to the end voltage at the autonomy time), temperature and ageing data | MISSING | [vendor to complete] | [vendor to complete] | [vendor to complete] |
| battery.temperature_correction_factor | Temperature correction factor | Battery data sheet with the constant-power discharge table (W per cell to the end voltage at the autonomy time), temperature and ageing data | MISSING | [vendor to complete] | [vendor to complete] | [vendor to complete] |
| battery.constant_power_discharge_data | Constant-power discharge data reference | Battery data sheet with the constant-power discharge table (W per cell to the end voltage at the autonomy time), | MISSING | [vendor to complete] | [vendor to complete] | [vendor to complete] |

| | | | | | | |
|---|---|---|---|---|---|---|
| | | temperature and ageing data | | | | |

Model, manufacturer, country of manufacture, and the type-test reports for the offered design: [vendor to complete].

## 7. Compliance schedule

The tenderer shall state, for every standard in section 2 and every technical requirement in section 5, whether the offer complies. 'Complies' needs an evidence reference (type-test report, certificate, calculation, drawing or data sheet with revision). A statement without evidence is recorded as not demonstrated. The requirement references below are the numbered items of section 5 of this template; no clause of any standard is quoted.

| Standard named in section 2 | Edition offered | Complies / complies with comment / does not comply | Evidence reference | Comment or deviation |
|---|---|---|---|---|
| IEC 60896 series (stationary lead-acid), if lead-acid is selected | [vendor to complete] | [vendor to complete] | [vendor to complete] | [vendor to complete] |
| IEC 62619 (secondary lithium cells and batteries, safety), if lithium-ion is selected | [vendor to complete] | [vendor to complete] | [vendor to complete] | [vendor to complete] |
| IEC 62485-2 (safety requirements for stationary batteries) | [vendor to complete] | [vendor to complete] | [vendor to complete] | [vendor to complete] |
| IEEE 485 (sizing lead-acid batteries) as a method reference | [vendor to complete] | [vendor to complete] | [vendor to complete] | [vendor to complete] |
| Local requirements [project to add] | [vendor to complete] | [vendor to complete] | [vendor to complete] | [vendor to complete] |

| Requirement (section 5) | Complies / complies with comment / does not comply | Evidence reference | Comment or deviation |
|---|---|---|---|
| 5.1 | [vendor to complete] | [vendor to complete] | [vendor to complete] |
| 5.2 | [vendor to complete] | [vendor to complete] | [vendor to complete] |
| 5.3 | [vendor to complete] | [vendor to complete] | [vendor to complete] |
| 5.4 | [vendor to complete] | [vendor to complete] | [vendor to complete] |
| 5.5 | [vendor to complete] | [vendor to complete] | [vendor to complete] |

## 8. Inspection and quality requirements

8.1 Review of the cell datasheet, rack drawings and safety documentation.

8.2 Inspection of delivery condition, cell voltages and torque.

8.3 Routine tests on every unit as required by the applicable standard and this specification.

8.4 Type-test reports or evidence for the design, issued by an accredited or recognised laboratory [project to define acceptable evidence].

8.5 Inspection and test plan (ITP) approved before manufacture, naming witness and hold points.

8.6 Calibration certificates for test instruments.

## 9. Factory acceptance test (FAT) requirements

9.1 Cell and module routine tests per the standard.

9.2 Capacity and discharge tests on a sample [project to define].

9.3 A FAT procedure shall be submitted for approval before the test; the results shall be issued as a signed report, with deviations and their disposition recorded.

9.4 Acceptance criteria are taken from the applicable standard and this specification [project to define]; a test passed against a criterion not stated in advance is not accepted.

9.5 Equipment shall not be shipped until the FAT report is accepted or a concession is recorded and approved by the owner.

## 10. Site acceptance test (SAT) requirements

10.1 Inspection, polarity and connection checks; insulation and voltage checks; an acceptance discharge test to the approved procedure; temperature and ventilation checks.

10.2 A SAT procedure shall be submitted for approval before the tests. It states the test, the method, the instrument, the acceptance criterion and the witness, with the criteria taken from the applicable standard and this specification [project to define].

10.3 SAT is carried out on the installed equipment with its final connections, after the installation inspection and before commissioning loads are applied.

10.4 Results are recorded on signed sheets. Any deviation, retest and concession is listed with its disposition. A SAT is not passed while a deviation is open.

10.5 SAT results are compared with the FAT results; a significant difference is investigated and recorded.

## 11. Commissioning requirements

11.1 Installation checks, polarity, torque, insulation resistance.

11.2 Initial charge per the manufacturer.

11.3 Capacity or autonomy test with the UPS at the design load; records retained as the baseline.

11.4 Commissioning records shall name the test, the instrument and its calibration, the date and the person, and shall be retained with the verification register.

11.5 Integrated systems testing of the whole power chain (utility loss, generator start and transfer, UPS on battery, loss of one power path, return to normal) is carried out under a project procedure [project to define]. This equipment's commissioning is not complete until it has taken part in that test.

## 12. Submittals and documentation

| Document | When required | Status |
|---|---|---|
| Completed vendor data schedule (section 6) and compliance schedule (section 7) | With the tender | [vendor to complete] |
| Datasheets and general arrangement drawings | With the tender, final before manufacture | [vendor to complete] |
| Type-test reports and certificates for the offered design | With the tender | [vendor to complete] |
| Inspection and test plan (ITP) | Before manufacture | [vendor to complete] |
| FAT procedure and report | Before and after the FAT | [vendor to complete] |
| Installation instructions and SAT procedure | Before delivery | [vendor to complete] |
| Operation and maintenance manuals, as-built drawings and final data | Before commissioning; as-built at handover | [vendor to complete] |
| Recommended spare parts and special tools lists | With the tender and at handover | [vendor to complete] |
| Training plan and materials [project to define] | Before commissioning | [vendor to complete] |
| Warranty terms [project to define] | With the tender | [vendor to complete] |

## 13. Open items (NOT VERIFIED)

- All ratings in section 4 marked PRELIMINARY or UNVERIFIED are open until confirmed from manufacturer data and the project studies.
- The prospective fault level is NOT CALCULATED until the utility, transformer and generator data are supplied.
- Standards editions and local requirements are NOT VERIFIED.
- The vendor data schedule (section 6) and the compliance schedule (section 7) are blank until a tenderer completes them.
- FAT, SAT and commissioning acceptance criteria are [project to define].
- This template has not been reviewed by an independent engineer.